Type MS · The TCP/IP model: four layers and what each does

Answer 1 of 6 · Question 3 part b · 4 marks

2021 NOV · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

3(a)	Award **one** mark for correct network type Award **one** mark for correct usage model One mark for each column		2

Network type	Tick (□)	Usage model	Tick (□)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	□
Personal Area Network (PAN)	□		

Type MS · The TCP/IP model: four layers and what each does

Answer 1 of 6 · Question 3 part b · 4 marks

2021 NOV · Paper 01 · extract 2 of 5 · find part b in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)</td><td>Award one mark for each correct layer name up to a maximum of two
Award one mark per layer for a correct function associated with the correct layer up to a maximum of two

<table><tr><th>Layer</th><th>Function</th></tr><tr><td>Application</td><td>• Selects and uses the correct protocol to transmit data
• Interacts with the user</td></tr><tr><td>Transport</td><td>• Identifies the server port to use
• Identifies the client port to use
• Divides the data into packets
• Numbers the packets
• Adds the total number of packets
• Sets up communication between hosts / Establishes end to end communication
• Passes the packets to the network layer
• Checks the packets arrive at the destination
• Resends any packets that have not arrived</td></tr><tr><td>Network</td><td>• Adds the source/sender's IP address
• Adds the destination IP address
• Routes the packets</td></tr><tr><td>Data link</td><td>• Controls physical connections between pieces of hardware
• Adds MAC addresses to the packets
• Sends the packets on their way
• Adds headers and trailers</td></tr></table></td><td>Accept functions at the receiving end too</td><td>4</td></tr>
</table>

Type MS · The TCP/IP model: four layers and what each does

Answer 1 of 6 · Question 3 part b · 4 marks

2021 NOV · Paper 01 · extract 3 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	**The only correct answer is A** B *is not correct because* it is not a smartphone frequency band C *is not correct because* it is not a smartphone frequency band D *is not correct because* it is not a smartphone frequency band		1

Question Number	Answer	Additional Guidance	Mark
3(d)(i)	Award **one** mark from: • higher frequency has more waves per second/carries more data in the same time (1) • higher frequency gives greater bandwidth (1) • higher frequency is more stable/less prone to interference (1)		1
3(d)(ii)	Award up to **two** marks for a linked explanation such as: • gives Carlo faster responses to his communications/connections (1) allowing Carlo to move/download/share data in a shorter time (1) • gives Carlo more reliable communications/can access communications more widely/in more places (1) giving Carlo a better user experience (1) • gives Carlo more secure communications (1) improving his privacy/reducing other people's ability to intercept/spy on his communications (1)		2

Type MS · The TCP/IP model: four layers and what each does

Answer 1 of 6 · Question 3 part b · 4 marks

2021 NOV · Paper 01 · extract 4 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(i)	Award **one** mark from: • do not leave the smartphone on standby/turn off when not being used (1) • use (portable) solar cells/other renewable generation method (1) • unplug charger when not in use (1) • reduce screen brightness (1) • turn off WiFi/GPS/Bluetooth/location (when not in use) (1) • close apps that are not being used (1)		1

Question Number	Answer	Additional Guidance	Mark
3(e)(ii)	Award up to **two** marks for a linked explanation. Award **one** mark for the action • keep smartphone for longer (1) • repair/recycle smartphone if not working (1) • upgrade the operating system (1) Award **one** mark for the impact • less e-waste is generated (1) • manufacturing could be reduced (1)		2
		Total	**13**

Type MS · The TCP/IP model: four layers and what each does

Answer 1 of 6 · Question 3 part b · 4 marks

2021 NOV · Paper 01 · extract 5 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The TCP/IP model: four layers and what each does

Answer 2 of 6 · Question 1 part f.ii · 2 marks

2020 NOV · Paper 01 · extract 1 of 5 · find part f.ii in this mark scheme extract

1(a)	Any **two** from: • (Share) access to the Internet/WWW/broadband connection (1) • (Internal) communication (using email/instant messaging/calendars) (1) • Share files/data (1) • Share peripherals/printers / hardware (1) • Increases the amount of storage (since network storage devices can be attached) (1) • Saves money on licences (since network site licences for software are usually cheaper than buying a stand-alone licence for every machine) (1) • Centralised backup (1) • Centralised security (1)	• Award examples if mapped to a bullet point • Do not award the same bullet more than once • Do not award for install updates (in the stem)	2

Question Number	Answer	Additional Guidance	Mark
1(b) (i)	Any **two** from: • ~~Faster (data transfer) (1)~~ Not really true, its Greater bandwidth / more bits per second (1) • The connection does not get worse the further you are from the router / more reliable (1) • Connection does not get obstructed by walls, ceilings, and furniture (1) • More secure (1)	Accept reverse arguments for disadvantages of wireless.	2

Type MS · The TCP/IP model: four layers and what each does

Answer 2 of 6 · Question 1 part f.ii · 2 marks

2020 NOV · Paper 01 · extract 2 of 5 · find part f.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b) (ii)	Any **two** from: • Installation/maintenance is more complex (1) • Devices need to be physically connected (1) • Less portable / limited by length of cable (1) • Limited number of devices can be connected (1) • Some digital devices cannot use a wired connection (1) • Trip hazard (1)	Accept reverse arguments for advantages of wireless.	2

Question Number	Answer	Additional Guidance	Mark
1(c)	Award **two** marks for a linked description that addresses the individual mark points: The internet is a (global) network of networks/ connected devices (1) whereas the www is the collection of web pages/service accessed using the internet (1) / the www is resources located via URLs/domain names (1) The internet is the infrastructure (1) and the www is a service that runs on that infrastructure (1)		2

Question Number	Answer	Additional Guidance	Mark

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Type MS · The TCP/IP model: four layers and what each does

Answer 2 of 6 · Question 1 part f.ii · 2 marks

2020 NOV · Paper 01 · extract 3 of 5 · find part f.ii in this mark scheme extract

1(d)	Award **two** marks for a linked explanation such as: Running out of IPv4 addresses/the number of (internet) connected devices has grown (1) IPv6 addresses are long/longer than IPv4 / the number of possible addresses is large/will last much longer (1)		2

Question Number	Answer	Additional Guidance	Mark
1(e)(i)	Star		1

Question Number	Answer	Additional Guidance	Mark
1(e) (ii)	Any **one** from: • easy to connect/remove new nodes (1) • failure of one node/link does not affect the rest of the network (1) • easy to detect the failure of one node/link (1)		1

Question Number	Answer	Additional Guidance	Mark
1(e) (iii)	Any **one** from: • if central switch/hub fails then the whole network fails (1) • performance and number of nodes that can be added depends on capacity of central switch/hub (1) • can require more cable than some of the other topologies (1)		1

Question Number	Answer	Additional Guidance	Mark

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Type MS · The TCP/IP model: four layers and what each does

Answer 2 of 6 · Question 1 part f.ii · 2 marks

2020 NOV · Paper 01 · extract 4 of 5 · find part f.ii in this mark scheme extract

1(f)(i)	Any **two** from: • SMTP (1) • IMAP (1) • POP/POP3 (1)		2

Question Number	Answer	Additional Guidance	Mark
1(f)(ii)	Any **two** from: • passes the (reassembled) packets to the application layer (1) • check if all packets have arrived (1) • determine whether the contents are correct (1) • requests resending of lost or damaged packets (1) • reassembles packets in correct order/ into an email (1)	Do not accept receives from network layer (stem)	2

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · The TCP/IP model: four layers and what each does

Answer 2 of 6 · Question 1 part f.ii · 2 marks

2020 NOV · Paper 01 · extract 5 of 5 · find part f.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The TCP/IP model: four layers and what each does

Answer 3 of 6 · Question 2 part c.i · 1 mark

2021 JUN · Paper 01 · extract 1 of 5 · find part c.i in this mark scheme extract

Number			
2(a)	Award **one** mark from: • share peripherals/devices (e.g. printers, scanners) (1) • communicate (e.g. email, instant messaging, play games) (1) • share data (e.g. files/music/videos/backups on servers) (1) • deploy/update applications/software(1) • administrative purposes (e.g. remote access/desktop)		1

Question Number	Answer	Additional Guidance	Mark
2(b)	Award **one** mark from: • to allow (connected) machines to communicate (1) • to provide the rules of communication between (networked) devices (1)	Accept any other response indicating protocols enable communication between devices	1

Type MS · The TCP/IP model: four layers and what each does

Answer 3 of 6 · Question 2 part c.i · 1 mark

2021 JUN · Paper 01 · extract 2 of 5 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	**The only correct answer is A** ***B*** *is not correct because FTP is an application protocol, which is added in the application layer* ***C*** *is not correct because FTP is an application protocol, which is added in the application layer* ***D*** *is not correct because FTP is an application protocol, which is added in the application layer*		1
2(c)(ii)	Award **one** mark from: • Adds the source/destination IP address (1) • Establishes sockets (an IP address followed by a port number) (1) • Moves packets onto the next network node (1) • Strips the source/destination IP addresses (when it arrives at its destination) (1) • Passes the packet to the transport/next layer (when it arrives at its destination) (1) • Adds/removes packet headers (1)		1

Question Number	Answer	Additional Guidance	Mark
2(d)	Award **one** mark for each of: **Domain Name** / **IP Address** / **URL** (blank) / (blank) / ✓ (blank) / ✓ / (blank) ✓ / (blank) / (blank)		3

Type MS · The TCP/IP model: four layers and what each does

Answer 3 of 6 · Question 2 part c.i · 1 mark

2021 JUN · Paper 01 · extract 3 of 5 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)(i)	Bus		1
2(e)(ii)	Award **one** mark from: • Easy to connect/remove devices/computers (in a network) (1) • Cable cost less / fewer cables needed (compared to mesh/star topology) (1) • Easier to plan/setup/maintain (1)		1
2(e)(iii)	Award **one** mark from: • If the main cable/a terminator fails or gets damaged the whole network will fail (1) • The more workstations the slower it gets (1) • The more workstations the more collisions (1) • Every device 'sees' all of the data on the network (1)		1

Question Number	Answer	Additional Guidance	Mark
2(f)(i)	Award **two** marks from: • Reduces time taken to upload/download (1) • Reduces the bandwidth required/cost of data to upload/download (1) • (Usually) produces smaller file/reduces storage (on the web server/device) (1) • People have a limited hearing range / can't tell the difference (1) • People's brains will 'fill in the blanks' (1)		2
2(f)(ii)	**The only correct answer is C** ***A*** *is not correct because it is not compressed* ***B*** *is not correct because it uses lossless compression* ***D*** *is not correct because it uses lossless compression*		1
2(f)(iii)	Award **one** mark from: • 3c3w2c6w3c (1) • c3w3c2w6c3 (1)		1

Type MS · The TCP/IP model: four layers and what each does

Answer 3 of 6 · Question 2 part c.i · 1 mark

2021 JUN · Paper 01 · extract 4 of 5 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(g)	Award **four** marks for a correct expression or correct answer **Examples of correct expressions:** 6 x 1000 x 1000 x 1000 x 8 (4) 54 x 1000 x 1000 1000 x 8 (4) 9 888.9 (4) For an incorrect or partial expression award **one** mark for each of: • **Bits to transfer** ○ 6×1000^3 (1) ○ x 8 (or equivalent) (1) • **Speed in bits per second** ○ 54×1000^2 (1) • Their bits divided by their speed (1)	• Units are not required • Equivalent expressions are accepted • Allow follow through for incorrect or partial expressions	4

Type MS · The TCP/IP model: four layers and what each does

Answer 3 of 6 · Question 2 part c.i · 1 mark

2021 JUN · Paper 01 · extract 5 of 5 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The TCP/IP model: four layers and what each does

Answer 4 of 6 · Question 2 part c.ii · 1 mark

2021 JUN · Paper 01 · extract 1 of 5 · find part c.ii in this mark scheme extract

Number			
2(a)	Award **one** mark from: • share peripherals/devices (e.g. printers, scanners) (1) • communicate (e.g. email, instant messaging, play games) (1) • share data (e.g. files/music/videos/backups on servers) (1) • deploy/update applications/software(1) • administrative purposes (e.g. remote access/desktop)		1

Question Number	Answer	Additional Guidance	Mark
2(b)	Award **one** mark from: • to allow (connected) machines to communicate (1) • to provide the rules of communication between (networked) devices (1)	Accept any other response indicating protocols enable communication between devices	1

Type MS · The TCP/IP model: four layers and what each does

Answer 4 of 6 · Question 2 part c.ii · 1 mark

2021 JUN · Paper 01 · extract 2 of 5 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	**The only correct answer is A** ***B*** *is not correct because FTP is an application protocol, which is added in the application layer* ***C*** *is not correct because FTP is an application protocol, which is added in the application layer* ***D*** *is not correct because FTP is an application protocol, which is added in the application layer*		1
2(c)(ii)	Award **one** mark from: • Adds the source/destination IP address (1) • Establishes sockets (an IP address followed by a port number) (1) • Moves packets onto the next network node (1) • Strips the source/destination IP addresses (when it arrives at its destination) (1) • Passes the packet to the transport/next layer (when it arrives at its destination) (1) • Adds/removes packet headers (1)		1

Question Number	Answer	Additional Guidance	Mark
2(d)	Award **one** mark for each of: **Domain Name** / **IP Address** / **URL** (blank) / (blank) / ✓ (blank) / ✓ / (blank) ✓ / (blank) / (blank)		3

Type MS · The TCP/IP model: four layers and what each does

Answer 4 of 6 · Question 2 part c.ii · 1 mark

2021 JUN · Paper 01 · extract 3 of 5 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)(i)	Bus		1
2(e)(ii)	Award **one** mark from: • Easy to connect/remove devices/computers (in a network) (1) • Cable cost less / fewer cables needed (compared to mesh/star topology) (1) • Easier to plan/setup/maintain (1)		1
2(e)(iii)	Award **one** mark from: • If the main cable/a terminator fails or gets damaged the whole network will fail (1) • The more workstations the slower it gets (1) • The more workstations the more collisions (1) • Every device 'sees' all of the data on the network (1)		1

Question Number	Answer	Additional Guidance	Mark
2(f)(i)	Award **two** marks from: • Reduces time taken to upload/download (1) • Reduces the bandwidth required/cost of data to upload/download (1) • (Usually) produces smaller file/reduces storage (on the web server/device) (1) • People have a limited hearing range / can't tell the difference (1) • People's brains will 'fill in the blanks' (1)		2
2(f)(ii)	**The only correct answer is C** ***A** is not correct because it is not compressed* ***B** is not correct because it uses lossless compression* ***D** is not correct because it uses lossless compression*		1
2(f)(iii)	Award **one** mark from: • 3c3w2c6w3c (1) • c3w3c2w6c3 (1)		1

Type MS · The TCP/IP model: four layers and what each does

Answer 4 of 6 · Question 2 part c.ii · 1 mark

2021 JUN · Paper 01 · extract 4 of 5 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(g)	Award **four** marks for a correct expression or correct answer **Examples of correct expressions:** 6 x 1000 x 1000 x 1000 x 8 (4) 54 x 1000 x 1000 1000 x 8 (4) 9 888.9 (4) For an incorrect or partial expression award **one** mark for each of: • **Bits to transfer** ○ 6×1000^3 (1) ○ x 8 (or equivalent) (1) • **Speed in bits per second** ○ 54×1000^2 (1) • Their bits divided by their speed (1)	• Units are not required • Equivalent expressions are accepted • Allow follow through for incorrect or partial expressions	4

Type MS · The TCP/IP model: four layers and what each does

Answer 4 of 6 · Question 2 part c.ii · 1 mark

2021 JUN · Paper 01 · extract 5 of 5 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The TCP/IP model: four layers and what each does

Answer 5 of 6 · Question 1 part d · 1 mark

2024 JUN · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

Number			
1(a)	Award **one** mark for any of the following up to a maximum of **two** marks: • WANs cover large/wide (geographic) distances/areas / LANs cover small/local (geographic) distances/areas (1) • WANs have lower bandwidth / LANs have higher bandwidth (1) • WANs connect multiple LANs / LANs are single networks (1) • WANs have higher latency / LANs have lower latency (1) • WANs use shared infrastructure / LANs are usually privately owned (1) • WANs use third party connections / LANs are owned by the organisation (1) • WANs have more users/devices / LANs have fewer users (1)	Do not award 2 marks for the same point e.g. A WAN is used over a large geographic area whereas a LAN covers a small distance. Two different differences must be identified. Do not accept general points such as WANs are slower - too vague. Do not accept LANs can be wireless.	2
1(b)	Award **one** mark for any of the following: • (Increased risk of) malware/viruses **spreading** / Pharming / Cyber attack (1) • (Increased risk of) hacking/unauthorised access/interception (1)		1
1(c)(i)	Award **one** mark for: • (Partially connected) mesh (1)		1

Type MS · The TCP/IP model: four layers and what each does

Answer 5 of 6 · Question 1 part d · 1 mark

2024 JUN · Paper 01 · extract 2 of 3 · find part d in this mark scheme extract

1c(ii)	Award **one** mark for any of the following: • More expensive to install (1) • Needs more cabling/wiring (1) • Requires more maintenance (1) • Harder to maintain / manage / install (1) • Higher power consumption (1)		1
1(c)(iii)	Award up to **two** marks for a linked explanation such as: • In the event of a damaged/congested connection data/packets can still be routed to their destination (1) because each node/computer is connected to at least two other nodes / there are multiple pathways available (1)		2
1(d)	**The only correct answer is D** *A is not correct because the application layer deals with applications* *B is not correct because the transport layer deals with packet structure* *C is not correct because the network layer deals with routing end to end*		1
1(e)(i)	Award **one** mark for: • https/hypertext transfer protocol secure (1)		1
1(e)(ii)	Award **one** mark for: • specification(.html) (1)		1
		Total for question 1	**10**

Type MS · The TCP/IP model: four layers and what each does

Answer 5 of 6 · Question 1 part d · 1 mark

2024 JUN · Paper 01 · extract 3 of 3 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The TCP/IP model: four layers and what each does

Answer 6 of 6 · Question 1 part b · 2 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

<table>
<tr><td>1(a)</td><td>Award one mark for each correct row:

<table><tr><th>Statement</th><th>DNS</th><th>PAN</th><th>WAN</th><th>WWW</th></tr><tr><td>A network of networks not limited to a single location</td><td></td><td></td><td>✓</td><td></td></tr><tr><td>A network that connects devices within the range of a single user</td><td></td><td>✓</td><td></td><td></td></tr><tr><td>A set of hypertext documents accessed through the HTTP protocol</td><td></td><td></td><td></td><td>✓</td></tr><tr><td>A system that maps domain names to IP addresses</td><td>✓</td><td></td><td></td><td></td></tr></table></td><td>Do not award a mark for a row containing more than one tick.</td><td>4</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(b)</td><td>Award one mark for each of:

• Transport (layer) (1)
• (Data) link / network link / network interface (layer) (1)</td><td>Accept
• transfer or transmission for Transport

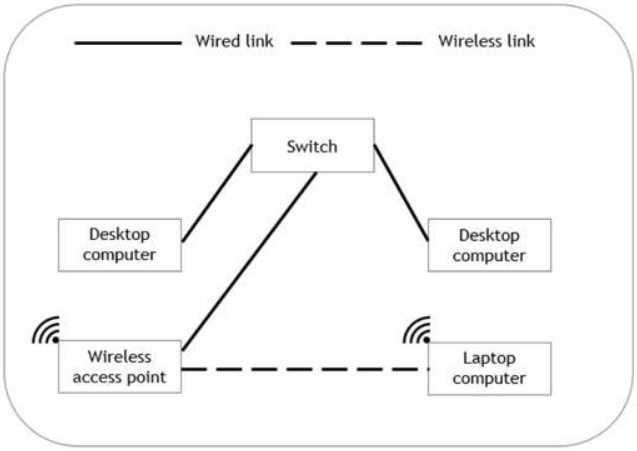
Do not accept:
• Network layer and Application layer (in the question)
• Data access layer, Internet layer (alternative names for network layer)
• Data’ or ‘network’ without ‘link’ or ‘interface’</td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(c)</td><td>The only correct answer is D

A is not correct because HTTP is used to transfer hypertext
B is not correct because it is used to view email held on a server
C is not correct because it is used to download email from a server</td><td></td><td>1</td></tr>
</table>

Type MS · The TCP/IP model: four layers and what each does

Answer 6 of 6 · Question 1 part b · 2 marks

2025 JUN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

Question	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each correct connection: • One solid line from wireless access point to switch (1) • One solid line from each desktop to switch (1) • One dashed line from laptop computer to wireless access point (1) 	No marks should be awarded if any additional lines have been drawn for each mark point. Do not double penalise (e.g. if you have penalised a second line from desktop to WAP then don't penalise extra line from WAP to desktop)	3

Type MS · The TCP/IP model: four layers and what each does

Answer 6 of 6 · Question 1 part b · 2 marks

2025 JUN · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

Question	Answer	Additional Guidance	Mark
1(e)	Award up to **two** marks for a linked description such as: • To find weaknesses in a system (1) fix/repair weakness (1) • To find vulnerabilities in a system (1) by testing the effectiveness of the security measures in place (1) • An ethical hacking attack on a network (1) to identify vulnerabilities / recommend measures/take action to improve security (1) • Tests whether users of the network (1) can be tricked into giving away information / giving criminals access to the network / downloading malware (1) • To test the system security (1) / sanctioned by the owner (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
		Total for question 1	**12**

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Type MS · The TCP/IP model: four layers and what each does

Answer 6 of 6 · Question 1 part b · 2 marks

2025 JUN · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

Question	Answer	Additional Guidance	Mark